

EDA Snacks – Problem Statement

Machine Learning Project

1. Background

The snack industry generates data related to products, prices, discounts, ratings, categories, locations, and sales. Analyzing this data can reveal patterns in customer purchasing behavior and help businesses understand factors associated with snack demand.

2. Problem Statement

The objective of this project is to perform Exploratory Data Analysis (EDA) on a snack sales dataset and use the insights obtained from the data to develop a Machine Learning model for predicting snack sales or demand. The project will investigate how variables such as price, discount, category, rating, location, season, and other available attributes are related to sales performance.

3. Objectives

- Understand the structure and characteristics of the snack dataset.
- Identify and handle missing, duplicate, inconsistent, and invalid data.
- Analyze distributions, trends, relationships, and correlations among variables.
- Detect and appropriately treat significant outliers.
- Engineer useful features for Machine Learning.
- Identify variables useful for predicting snack sales or demand.
- Build and evaluate suitable Machine Learning models.

4. Proposed ML Task

The primary proposed task is regression: predicting the quantity of snacks sold using available product and sales-related features. If the dataset is better suited to categorical demand levels, the task can instead be formulated as classification into Low, Medium, and High demand.

5. Expected Outcome

The project is expected to produce a cleaned and analyzed snack dataset, meaningful EDA visualizations and findings, a set of ML-ready features, and an evaluated predictive model. The results should demonstrate how EDA supports the Machine Learning workflow.

6. Scope

The project covers data preprocessing, exploratory analysis, feature engineering, model development, and evaluation. Advanced deployment, real-time recommendation systems, and production-scale infrastructure are outside the scope of the initial project.